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Supplementary appendix

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Supplement to: Imazio M, Belli R, Brucato A. Efficacy and safety of colchicine for treatment of multiple recurrences of pericarditis (CORP-2): a multicentre, double-blind, placebo-controlled, randomised trial. *Lancet* 2014; published online March 30. [http://dx.doi.org/10.1016/S0140-6736\(13\)62709-9](http://dx.doi.org/10.1016/S0140-6736(13)62709-9).

CORP 2 (COLchicine for Recurrent Pericarditis) – a randomized placebo-controlled trials evaluating the clinical benefits of colchicine as adjunct to conventional therapy in the treatment and prevention of multiple recurrences of pericarditis.

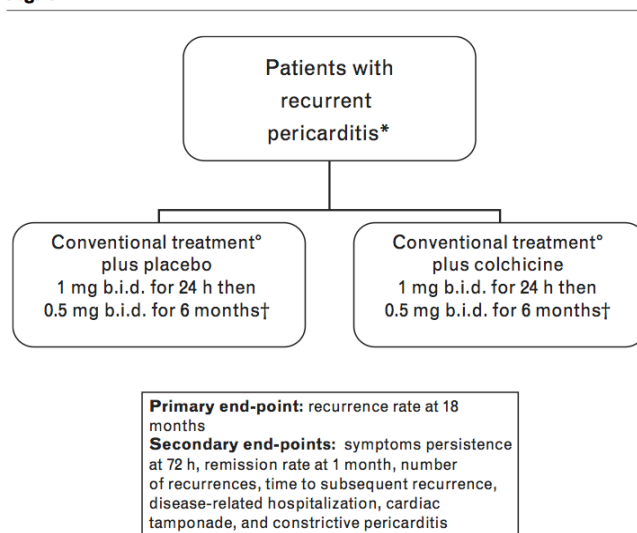
Introduction

Colchicine has been used for hundreds of years as an anti-inflammatory agent for acute arthritis and is the most specific known treatment for acute attacks of gout [1 – 3]. More recently, the drug has been successfully used for the prophylaxis of familial Mediterranean fever attacks [4,5] and the treatment of recurrent pericarditis [6 – 15]. Evidence for the possible utility of colchicine in the treatment and prevention of recurrent pericarditis comes from case reports and non-randomized, observational studies. On this basis, the 2004 European guidelines [16] have recommended the use of colchicine in the treatment of recurrent pericarditis. A preliminary open- label trial (CORE trial) [17] reported a possible utility of colchicine as adjunct to conventional therapy in the treatment of the first recurrence. However, several authors do not recommend the use of the drug in the first recurrence but only after failure of conventional therapies in patients with two or more relapses [18]. The same authoritative authors have underlined that colchicine is able to control symptoms in some patients, whereas it can diminish the severity or the frequency of the relapses in others but, unfortunately, the drug is not effective in all cases. According to their experience, colchicine appears to be useful in more than one-half of patients with two or more relapses. The precise proportion of responders is not exactly known because appropriately designed placebo controlled prospective studies are lacking, and it is unfortunate that such a broad use of colchicine is based on rather weak evidence [18]. Unfortunately, multicentre placebo-controlled randomized trials are lacking to guide the evaluation and management of pericarditis [19,20], and colchicine might represent a cheap, safe option for the treatment and prevention of recurrent pericarditis [6,13,17,18]. To evaluate the possible utility of colchicine in multiple recurrences, we planned the CORP-2 trial.

Study objectives

This prospective, randomized, double-blind, placebo- controlled, multicentre trial will evaluate the efficacy and safety of colchicine in the treatment and prevention of multiple recurrences of pericarditis. Colchicine will be considered as adjunct to conventional treatment. The trial design is reported in Fig. 1.

Fig. 1



Methods

Study population

The study population will consist of 240 patients with a second or subsequent relapse. Patients will be enrolled in Italy.

Inclusion criteria

All consecutive patients with the second or subsequent relapse will be enrolled in the CORP-2 trial. Eligible patients will have no contraindication to colchicine, will be able to provide their informed consent, and have no unfavourable short-term outlook. Inclusion criteria will be a definite diagnosis of recurrent pericarditis (idiopathic, viral and autoimmune aetiologies, including postpericardiotomy syndromes and connective tissue diseases), age >18 years, and provision of informed consent.

Recurrent pericarditis will be diagnosed either by recurrent or continued activity of pericarditis [6,17,21] after a documented prior episode of acute pericarditis meeting accepted diagnostic criteria [22 – 27] (Table 1).

Table 1 Diagnostic criteria for acute and recurrent pericarditis

Disease	Criteria
Acute pericarditis	At least 2 out of 4: 1. Typical chest pain 2. Pericardial friction rub 3. Widespread ST segment elevation or PR depressions 4. New or worsening pericardial effusion
Recurrent pericarditis	Previous history of acute pericarditis (definite diagnosis) plus recurrent pain and at least one of the following criteria: 1. Fever 2. Pericardial friction rub 3. Electrocardiographic changes 4. New or worsening pericardial effusion 5. Elevations in the white blood cell count, erythrocyte sedimentation rate or C-reactive protein

Recurrence will be documented by recurrent pericarditic pain and one or more of the following signs: fever, pericardial friction rub, electrocardiogram changes, echocardiographic evidence of new or worsening pericardial effusion, and elevations in white blood cell count, erythrocyte sedimentation rate or C-reactive protein.

Exclusion criteria

Patients meeting any of the following exclusion criteria will not be eligible for the study: first attack of acute pericarditis, tuberculous, neoplastic or purulent aetiologies, known severe liver disease or current transaminases >1.5-fold higher than the upper normal limit, current

serum creatinine above 2.5mg/dl, known myopathy or current serum creatine kinase above the upper normal limit, known blood dyscrasias or gastrointestinal disease, pregnant and lactating women or women of childbearing potential not protected by a contraception method, known hypersensitivity to colchicine, and current treatment with colchicine for any indications.

Study design

Study drug administration. Patients will be randomized to receive placebo or colchicine on top of a conventional treatment with aspirin 800mg orally every 6–8h for 7 – 10 days with gradual tapering over 3 – 4 weeks. Colchicine will be given at the dose of 1.0 – 2.0 mg for the first day followed by a maintenance dose of 0.5 – 1.0 mg daily for 6 months. The lower dose (initial dose: 0.5mg b.i.d. and maintenance dose 0.5mg daily) will be given to patients who weigh less than 70 kg or who are intolerant to the highest dose (initial dose 1.0 mg b.i.d. and maintenance dose of 0.5 mg b.i.d.). As the preferred non-steroidal anti-inflammatory drug (NSAID), we will use aspirin according to previous published experience [17,21,24,25]. Based on previous experience [17,28], it has been decided to assign patients to the lowest effective dose of colchicine, thus reducing side-effects and improving drug tolerability. Corticosteroid therapy (prednisone at the dose of 1.0 mg/kg/day for 4 weeks and then gradually tapered) will be restricted to patients with aspirin contraindications (allergy, history of peptic ulcer or gastrointestinal bleeding, and oral anticoagulant therapy when the bleeding risk is considered to be high or unacceptable) or intolerance. In every case, a gastroduodenal prophylaxis will be adopted using a proton pump inhibitor also without initial evidence of gastrointestinal intolerance [16,17,20,21,24,25,28].

Concomitant medications. Additional medical therapy will be adopted according to comorbidities, existing practice guidelines and local practice.

Hypotheses

The primary hypothesis of the CORP-2 trial is that, in patients with recurrent pericarditis, the use of colchicine as adjunct to conventional treatment, will halve the recurrence rate at 18months. The secondary hypothesis includes colchicine reducing symptom persistence at 72h, as well as the total number of recurrences, disease-related hospitalization, frequency of cardiac tamponade and pericardial constriction, and increasing the remission rate at 1 week and the time to the subsequent recurrence.

End-points

The primary study end-point will be the recurrence rate at 18 months. The secondary end-points are symptom persistence at 72 h, remission rate at 1 week, number of recurrences, the time to first recurrence, disease-related hospitalization, cardiac tamponade and constrictive pericarditis rates. A blinded clinical end- point committee will adjudicate all end-points.

Diagnostic criteria for recurrence are reported in Table 1. Patients will be considered to be in remission when they are symptom-free, with disappearance of clinical, electrocardiographical and echocardiographical signs [10,17,21,25,28]. On the contrary, treatment failure will be considered in case of an unfavourable clinical reaction with persistence of fever, pericardial effusion appearance or worsening, and general illness lasting more than 7 days [6,10,17,18,21,25,28].

Sample size and data analysis

For the CORP-2 trial, a total of 240 patients, 120 in each treatment arm, will be needed to detect a difference in the recurrence rate of 30.0% and 15.0% between the two treatment arms (placebo and colchicine) with a power of 80% ($P < 0.05$, two-sided). The estimated rate of recurrent pericarditis of 30.0% in the placebo group is based on the average recurrence rate [13,18]. Analyses will be performed by intention-to-treat.

A detailed summary of the study main features is reported in Table 2.

Data management

Data will be collected using case report forms, clinical events committee adjudication forms and a study database. Data will be managed blind to treatment assignments. The only individuals with access to unblinded event rates before locking the database will be members of the Safety and Clinical Events Committee.

Table 2.

Feature	CORP-2 trial
Study design	Randomized, double-blind, placebo controlled
Eligibility	Second or subsequent recurrence
Primary outcome	Recurrence rate at 18 months
Secondary outcomes	Symptom persistence at 72 h, remission rate at 1 week, number of recurrences, time to recurrence, disease-related hospitalization, cardiac tamponade, and constrictive pericarditis
Study treatment	Colchicine/placebo
Treatment length	6 months
Expected total enrollment	240

Study organization

Steering committee. The Steering Committee comprises a geographically diverse group of academicians and clinicians. This group will provide clinical guidance on protocol development, study implementation and conduct, and interpretation of results.

Safety and Clinical Events Committee. The committee comprises physicians with expertise in pericardial disease management, who will monitor the trial progress, review data from one interim efficacy analysis, and ensure the safety of patients being enrolled in the study. Safety data will be reviewed after the following numbers of patients have been enrolled: 40, 80, 120, 160 and 200.

The committee, blind to treatment assignment, will systematically identify and adjudicate suspected end-point events according to prespecified criteria. Adjudicated data will be used in the final efficacy and safety analyses.

Discussion

The CORP-2 trial is designed to assess the efficacy and safety of colchicine for the treatment of multiple recurrences of pericarditis, and the secondary prevention of further recurrences. Preliminary experience has shown that the early use of the drug might be beneficial even at the first recurrence [17]. In the CORE trial, a total of 84 patients with recurrent pericarditis (first episode) were randomly assigned to conventional treatment with aspirin or conventional treatment plus colchicine. After 1682 patient-months follow-up, colchicine reduced the recurrence rate at 18 months (24.0% versus 50.6%, respectively; $P=0.02$). Colchicine was discontinued in 7.0% of cases because of gastrointestinal intolerance, but no serious adverse effect was reported.

Despite these encouraging results, the data can be considered only as preliminary and require further validation in large multicentre placebo-controlled randomized trials before colchicine can be routinely recommended. The observed benefit is mainly due to the possible secondary prevention of recurrent pericarditis.

Recurrence is a common complication after acute and recurrent pericarditis. The exact recurrence rate is unknown, but has been reported to be 15 – 30% [6,16,18 – 20] and even up to 50% [21]. A reasonable estimate may be 30%. After a first recurrence, the expected recurrence rate may be as high as 50% according to the CORE trial [17]. Recurrent pericarditis can be a prolonged and frustrating disease with disabling pain and malaise. In some instances, symptoms may only be controlled by a corticosteroid or other immunosuppressive drugs. Although colchicine has been recommended in the European guidelines [16], this indication was based on the consensus of the experts and results obtained from case reports and non-randomized studies on recurrent pericarditis. However, no double-blind, randomized trials are available to support this advice.

Therefore, the purpose of the CORP-2 trials is to assess whether the addition of colchicine on top of established therapies can reduce the recurrence rate in recurrent pericarditis. Colchicine will be started using weight-adjusted doses to reduce any possible gastrointestinal side-effects that may be responsible for drug withdrawal.

Conclusion

The CORP-2 trial will evaluate the safety and efficacy of colchicine for the secondary prevention of recurrent pericarditis, respectively, in the second or subsequent recurrence. This study will also provide important information on the frequency, clinical presentation and prognosis of recurrent pericarditis in clinical practice.

Appendix

Steering Committee: Massimo Imazio, MD and Rita Trinchero, MD (Chairs), and nucleus members of the Study Group on Heart and Infectious Diseases of the Associazione Nazionale Medici Cardiologi Ospedalieri (ANMCO).

Safety and Clinical Events Committee: Yehuda Adler, MD (Tal Hashomer, Israel), David H Spodick, MD (Worcester, Massachusetts, USA), and Ralph Shabetai, MD (San Diego, California, USA- deceased).

Recruiting Centers and Investigators:

Cardiology Department, Maria Vittoria Hospital, Torino, Italy (Coordinating Center): Massimo Imazio (Principal Study Investigator), Riccardo Belli, Federico Beqaraj, Daniela Demarie, Silvia Ferro, Davide Forno.

Department of Internal Medicine, Ospedali Riuniti, Bergamo, Italy: Antonio Brucato, Silvia Maestroni, Davide Cumetti.

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Cardiology Department, Ospedale degli Infermi, Rivoli, Italy: Stefania Ferrua, Ferdinando Varbella.

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Summary of original protocol changes (before starting of the randomization):

1. Use of colchicine without loading dose (to reduce gastrointestinal side effects and improve patient compliance):

Colchicine will be given at a dose of 0.5 to 1.0 mg daily for six months without a loading dose to reduce potential gastrointestinal side effects and improve patient compliance. The duration of colchicine therapy is based on prior studies. The lower dose (0.5 mg daily) will be given to patients weighing 70 kg or less and to those intolerant of the higher dose (0.5 mg twice daily). Colchicine tablets will contain 1 mg of the active drug. All tablets (colchicine and placebo) will be identical in color, shape, and taste and premarked to allow splitting into two equal parts. Adherence to study drug therapy will be assessed on the basis of counts of pills in dispensed boxes, with a target of at least 80% adherence.

2. Concomitant anti-inflammatory drug will include aspirin, ibuprofen, and indomethacin (according to treating physician choice):

All patients also will receive conventional treatment for recurrent pericarditis based on three possible non-steroidal anti-inflammatory drugs (aspirin, ibuprofen, and indomethacin). Possible treatments will be either aspirin 800 mg or ibuprofen 600 mg or indomethacin 50 mg orally every 8 hours for 7 to 10 days with tapering over 3 to 4 weeks.

3. Prescription of low to moderate doses of prednisone instead of high doses (1.0 mg/kg/day):

Corticosteroid therapy (prednisone 0.2 to 0.5 mg/kg/day for 2 weeks, then gradually tapered) will be used in patients with contraindications to aspirin, ibuprofen, and indomethacin (allergy, history of peptic ulcer or gastrointestinal bleeding, or use of oral anticoagulant therapy when the bleeding risk was considered high or unacceptable) or a history of intolerance. All patients will receive a proton pump inhibitor for gastroduodenal prophylaxis.

Data source for the rationale of prednisone doses change: Imazio M, Brucato A, Cumetti D, et al. Corticosteroids for recurrent pericarditis: high versus low doses: a nonrandomized observation. Circulation. 2008;118:667-71.